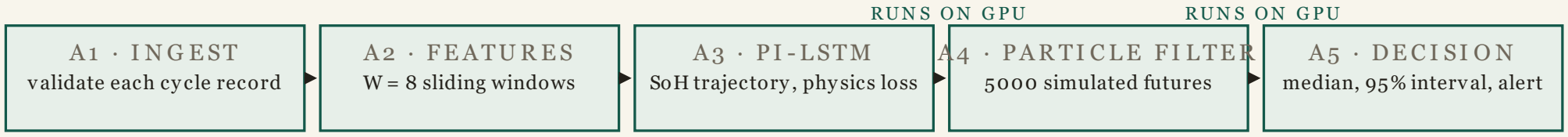
# Parallel Physics-Informed Prognostics for **EV Battery Health Management**

*A. Vibhas Chowdhary · Maddala Jithendra — guide: Dr. Kavitha R*

Department of Computing Technologies, SRM IST, Kattankulathur · va1641@srmist.edu.in · jm4408@srmist.edu.in · code, paper and full validation report: github.com/Jithendra123892/p3-evhm (MIT) · GPU runs: kaggle.com/code/jithendramaddala/p3-evhm-gpu-run




input: cycle capacity q(t) and mean temperature T(t) from NASA PCoE cells B0005 / B0006 / B0007 / B0018 · output: SoH forecast, RUL median + 95% interval,

The five-agent pipeline runs as one installable package (`pip install -e .`), a CLI benchmark, a six-tab live dashboard and a one-click Kaggle GPU kernel — the same code, four doors.

## THE PROBLEM / WHY BATTERY HEALTH IS HARD

A lithium-ion cell loses capacity with every charge cycle — slowly at first, then through a degradation *knee* that takes it below the 70–80% end-of-life threshold with almost no warning. A vehicle reports a charge percentage, not health, so the first sign of trouble is often the failure itself. What maintenance actually needs is a number with an error bar, enough cycles in advance to schedule the replacement.

Each ingredient of a solution exists in the literature — physics-informed training losses, LSTM × particle-filter hybrids, GPU-resident particle filters — but no published framework combines them on a real benchmark; the closest hybrid paper lists our physics constraints as *future work*. P3-EVHM fuses the four and measures what the fusion buys.

## METHOD / WHAT THE MODEL IS TOLD NOT TO DO

A 1–32 LSTM learns capacity dynamics from a W = 8 cycle window, but its loss carries two laws of electrochemistry: the prediction **may never claim the battery gained capacity**, and its decline must respect an Arrhenius-style temperature prior fitted from the training cycles themselves.

(4) $L = MSE(\hat{q}, q) + \lambda_1 \cdot mean\ ReLU(\hat{q}_t - q_t) + \lambda_2 \cdot mean(\Delta\hat{q}_t - g(T_t))^2$

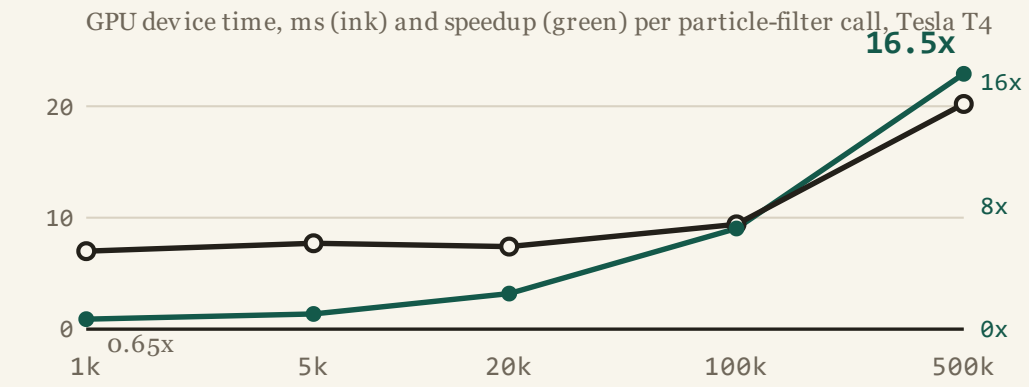
(5) $g(T) = clamp(\hat{A} \cdot (T - \bar{T}), min = 0)$ — *a prior, not a hard clamp: violations fall ~5× but not to zero*

(6–8) $q_i^{(k)} = q_i^{(k-1)} - d_i + \varepsilon_k;\ \ w_i \propto exp(-(q_{obs} - q_i)^2 / 2\sigma^2);\ \ RUL = median\{ k : q_i^{(k)} \le q_{EOL} \}$

Prognosis is a distribution, not a point: the trained model seeds N simulated futures, each with its own decline rate, reweighted by the *measured* start-cycle capacity; the cycle at which a future crosses the EOL threshold is that future's RUL. The model reports the median and the 95% interval — and the interval is calibrated, not decorative.

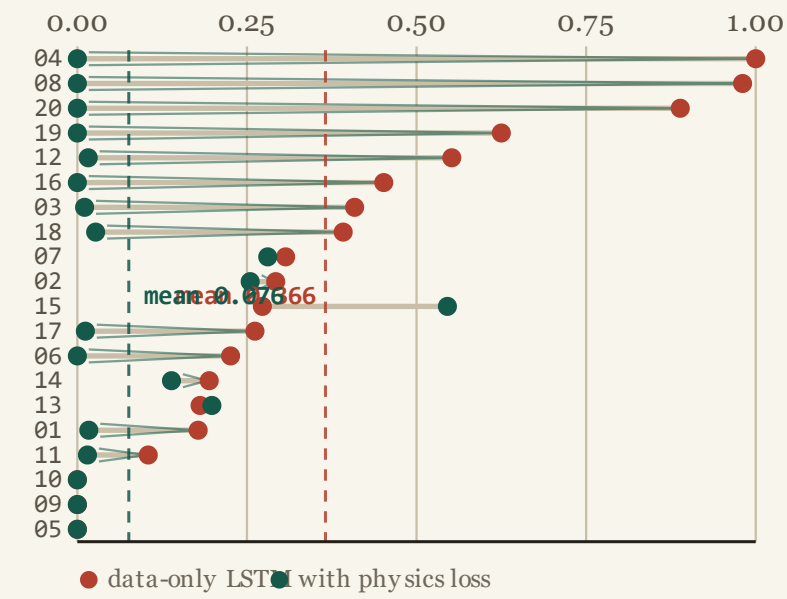
## PARALLELISM / ONE CLOUD, ONE TENSOR OP

Futures are independent, so the particle cloud is embarrassingly parallel: each propagation step is a single vectorised tensor update over all N particles — data-parallel SIMT on the GPU, the identical NumPy code on the CPU. No multiprocessing, no multi-node.




At small N the warm CPU wins — we report that. The GPU pays off exactly where particle-filter prognostics needs it: large clouds for sharp intervals, in real time.

## RESULT 1 / THE PHYSICS LOSS WORKS




All 20 grid configurations, Tesla T4, sorted by how badly the data-only LSTM behaves. Configs 04 and 08 are the catastrophes the physics loss eliminates (V = 1.0 → 0.0).

## RESULT 2 / WHAT WE MEASURED, ALL OF IT

| QUANTITY | MEASURED | NOTE |
|---|---|---|
| implausible steps V, mean | 0.366 → 0.076 | ~5× cut, 40-config grid, both machines |
| capacity MAE, mean | 0.285 → 0.257 Ah | physics loss also more accurate |
| RUL produced | 20 / 20 | physics+PF, every config, CPU *and* T4 |
| mean RUL error | 18.8 cyc | 18.6 on CPU — reproducible |
| 95% interval coverage | 0.85 | 17/20 — honest, below the 0.95 ideal[d] |
| GPU speedup at 500k | 16.5× | 20.2 ms device time per PF call |

## RESULT 3 / NEW CELL, NO RETRAINING

Leave-one-cell-out, 16 adaptations. A 21 KB fleet-pretrained checkpoint, body frozen, only the head fine-tuned:

| ADAPTATION MODE | RUL AVAIL. | V MAX | TIME |
|---|---|---|---|
| train from scratch (200 ep) | 1 of 4 | 0.355 | 2.85 s |
| pretrained, zero-shot | 4 of 4 | 0.000 | 0 s |
| head fine-tune (frozen body) | 4 of 4 | 0.000 | 0.20 s |

From-scratch wins raw capacity MAE on 3 of 4 cells — but fails the actual prognostic task. Transfer buys RUL in every held-out cell, zero physics violations, 14× faster.

## HONEST LIMITS / WHAT THIS IS NOT

Laboratory single-cell data — pack-level validity untested[d]. Early prediction starts (40% of life, before the knee) can fail without the decline-prior floor[d]. LSTM seed variance is material and reported. A research decision-support prototype — not a certified BMS safety function.

**Key references** — [1] Saha & Goebel, *Battery Data Set*, NASA Ames PCoE, 2007. [2] Arulampalam et al., *IEEE Trans. Signal Process.* 50(2), 2002. [13] Shi, Rivera & Wu, *Mech. Syst. Signal Process.* 179:109347, 2022. [17] *Parallel accelerated CRPF on CUDA*, Appl. Intell., 2020. [22] do Nascimento et al., *Sci. Rep.* 13, 2023. [28] *Hybrid PF-trained NN for Li-ion prognosis*, IEEE, 2021. Full 33-entry bibliography in the paper.

[d] documented in VALIDATION_REPORT.md with per-number provenance. Every figure on this poster is generated from `results/kernel_grid_stats.json` — nothing is hand-drawn or estimated.